Kuramoto Synchronized Configuration Space Trajectory Generation for an Ensemble of Manipulators Sharing a Common Workspace

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***Abstract*—We present an approach to generate collision-free synchronized configuration-space trajectories for an ensemble of serial-link manipulators sharing a common workspace, on a cell of six arms whose reachable volumes mutually overlap. We assume that per-arm task-space goals are given and that the relative timing among the arms is free. We pose the problem as one of temporal resolution first and geometric resolution only where timing is insufficient: each arm advances along an unmodified B-spline path under a Kuramoto phase-oscillator model, in which a repulsive coupling term brakes the trailing arm of a converging pair so that the leader passes first. When phase modulation alone cannot separate a pair, we deform only that pair's geometry using a perpendicular escape derived from the Jacobian transpose at the witness points of the deepest contact, applying equal and opposite displacements to both arms so that each deviates minimally from its nominal path. The escape is injected into the trajectory as B-spline control points, with local control-point refinement. Hence, the deformation remains confined to the contacting arc, after which the full synchronization sweep is re-run on the modified paths.**

***Index Terms*—B-spline trajectories, collision avoidance, manipulator ensembles, multi-arm coordination, phase synchronization, trajectory generation.**

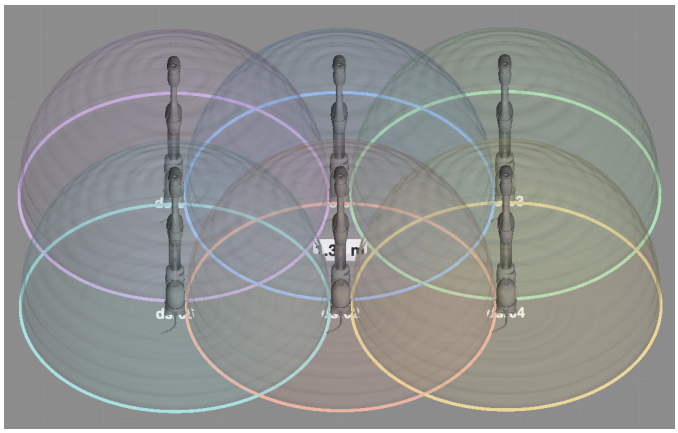


Fig. 1. The six-arm cell, rendered on the 1.39 m base grid with each arm's reachable volume shown as a translucent hemisphere and the 1.39 m nearest-neighbour spacing annotated. Every pair of neighbouring arms has overlapping reach; at this spacing interference is the normal operating condition rather than an exceptional one.

I. Introduction

Motion planning problems arise in assembly, machine tending, and increasingly in cells of multiple manipulators sharing a workspace. A cell of N arms placed to jointly service a common region necessarily has overlapping reachable volumes: the overlap is what makes the cell useful, and it is also where the arms interfere with one another. Fig. 1 shows the six-arm cell considered in this paper.

Two classes of methods address this coupling. Centralized planners search the composite configuration space $\mathcal{C} = \mathcal{C}_1 \times \cdots \times \mathcal{C}_N$, which is complete in principle but grows in dimension with N; for six 6-DOF arms this space is 36-dimensional. Decoupled methods plan each arm independently and reconcile the resulting trajectories afterward, either by assigning priorities and treating higher-priority arms as moving obstacles [3], or by fixing paths and adjusting only the timing along them [5], [6].

We observe that a decoupled reconciliation stage has two available instruments, retiming and deformation, and that they are not equally expensive. Each arm's independently planned trajectory already satisfies its own endpoint, kinematic, and actuator constraints. Retiming preserves the path exactly and only changes the schedule; deformation displaces the trajectory from the one the arm's own planner selected. A reconciliation stage that reaches for deformation before exhausting retiming pays this cost on every instance that timing alone could have resolved.

We realize the temporal instrument with a Kuramoto phase-oscillator model, adapted from its classical form in three respects developed in Section IV-E: the phase indexes progress along a finite path instead of a periodic cycle, the coupling is repulsive instead of attractive, and the coupling law is multiplicative instead of additive. The geometric instrument, used only where the temporal one fails, is a perpendicular escape derived from the witness points of the deepest contact and injected into the trajectory as B-spline control-point displacements.

Several of the individual mechanisms below are adaptations of established techniques – repulsive Kuramoto coupling [20], the Jacobian-transpose operational-space escape [10], and B-spline local support and knot insertion [11], [16] are each prior art and are cited at the point they are used. What is new is not any one of these pieces in isolation but their combination into a single two-tier loop, and the specific adaptations required to make each piece work in that loop. The contributions of this paper are as follows.

1) A two-tier formulation of multi-arm interference resolution in which phase synchronization is attempted before any geometric modification, with an observation (Section IV-A) characterizing exactly when timing alone suffices.
2) An adaptation of repulsive Kuramoto coupling [20] to

a finite, non-periodic progress variable, realized as a multiplicative rate law so that monotonicity holds by construction rather than by tuning a gain, together with a lag floor that provably guarantees termination and an ordering search whose cost scales with the number of conflicting pairs rather than the number of arms (Section IV-E).

3) A tight coupling between the two tiers: every geometric deformation (Section IV-F) is local by construction – confined to one pair and one contacting arc – but its effect on a third arm sharing a base is not, so after every accepted deformation the full Kuramoto sweep of Section IV-E is re-invoked rather than a targeted local check. This is what lets a repair confined to one pair be verified against, and if necessary absorb, interference it newly introduces with a different arm, without a global re-optimization over all trajectories. We show (Table III) that this re-invocation is not bookkeeping: it is what resolves the majority of interfering trials in every ensemble size tested.

II. Related Work

A. Multi-Robot Motion Planning and Path-Velocity Decomposition

Sampling-based planners such as PRM [1] and RRT-Connect [2] provide probabilistic completeness in a composite configuration space. Prioritized planning [3] treats higher-priority robots as moving obstacles for lower-priority ones – efficient but order-dependent. Conflict-based search [4] recovers completeness for multi-agent path finding on graphs via a two-level search. Our method is decoupled but does not commit to a fixed priority order: it searches pairwise phase orderings and invokes geometric repair only where temporal coordination fails. Kant and Zucker [5] separate path planning from time parameterization along a fixed path; O'Donnell and Lozano-Pérez [6] apply this to manipulator pairs via the coordination diagram, exact for two robots but growing in dimension with N. Our phase-oscillator formulation trades this exactness for scalability via local pairwise phase dynamics.

B. Trajectory Deformation

CHOMP [7], STOMP [8], and TrajOpt [9] optimize a full trajectory against collision and smoothness costs; TrajOpt certifies continuous-time safety via swept-volume checking and signed distance, informative even under penetration. Our geometric repair differs in scope: a local edit at a single contact, confined to two arms, accepted only on strict decrease of the global violation count, aiming for minimal disturbance to an already-optimal seed rather than global re-optimization. We evaluate discrete samples and an unsigned surface distance saturating at zero under penetration, with a witness-point direction recovered from a medial-axis fallback in that case; the escape direction follows the operational-space convention of mapping a Cartesian repulsion into joint space through the Jacobian transpose [10].

C. B-Spline Trajectory Representation and Local Modification

Splines are the standard parameterization for manipulator trajectories because they deliver bounded-derivative, continuous motion from a small number of parameters [11]–[13], and control points and time scaling are widely used as optimization variables for minimum-jerk and time-optimal generation [14], [15]. Two properties of the B-spline basis are central to our method. The first is *local support*: a basis function of degree p is non-zero over only $p + 1$ knot spans, so displacing a single control point perturbs the curve only locally, with continuity preserved automatically – what allows a collision repair to be a control-point displacement without re-solving the trajectory globally. The second is *knot insertion*, which increases control points without altering curve geometry [16]; refining the polygon locally shrinks each basis function's support and tightens the spatial extent of a subsequent edit.

Prior work has used spline representations primarily as the *decision variable* of a global optimization: Kabir et al. [24], [25] adaptively select the parametric representation of each configuration variable and solve for all parameters together under path constraints. We use the representation differently: control points are not optimization variables but the *actuator of a local geometric repair* (Section IV-F gives the mechanism), with endpoint control points pinned so the start pose and goal configuration remain exactly satisfied. Because the entire geometric-repair stage acts by displacing B-spline control points, we refer to it throughout this paper as *deformation*. Deformation and Kuramoto phase synchronization are the two mechanisms this paper combines, and Table III reports the fraction of interference resolved by each. The closest prior technique is the elastic band [23], also displacing path control points under obstacle repulsion; ours differs in triggering only at a single violated contact rather than a continuous force balance, confining the edit to the contacting arc by local knot insertion, and accepting it only on strict decrease of the violation count rather than as a continuous dynamic – and is unrelated to the classical *retraction method* of exact motion planning [22], which deforms a rigid body's free configuration *space* onto a generalized Voronoi diagram, not extended to high-DOF manipulators for the dimensionality reasons sampling-based planners [1], [2] are preferred there. What we deform is a single arm's *trajectory geometry* $\boldsymbol{\gamma}_i$ alone, applied once between synchronization sweeps, never touching $\phi_i(t)$, so an arm's progress along its (possibly deformed) path still only moves forward, per the monotonicity guarantee of Section IV-E.

D. Coupled Oscillators and the Kuramoto Model

Coupled-oscillator models are used in robotics chiefly as central pattern generators for periodic locomotion [17]. The Kuramoto model [18], [19] is the canonical formalism for phase synchronization and is reviewed in full in Section IV-E. Repulsive coupling is itself an established variant: Hong and Strogatz [20] study a Kuramoto population coupled to the mean field with either sign, so "contrarian" oscillators are repelled toward antiphase, characterizing the resulting partially-

synchronized and travelling-wave states. We adopt repulsive coupling for the same qualitative reason, but the setting differs in every other respect: existing applications, repulsive or attractive, synchronize *periodic* motion (Table I), whereas ours is a finite, point-to-point motion, desynchronizing arrival at a shared region rather than synchronizing any quantity. To our knowledge, applying repulsive Kuramoto coupling to a non-periodic, finite progress variable for manipulator interference resolution has not previously been reported.

E. Synchronized Trajectory Generation for Ensembles

Kabir et al. [24], [25] generate synchronized configuration-space trajectories for an ensemble of robots under workspace path constraints, posing trajectory generation as a discrete parameter optimization over B-spline control points with successive constraint refinement. There are relative motion among objects is a given path constraint, and the difficulty is satisfying it jointly across all robots. In our problem no relative motion constraint is imposed among the arms; each performs an independent task in a shared volume, and relative timing is a free variable for interference resolution rather than a quantity the task constrains.

III. Problem Formulation

Let $\mathcal{R} = \{R_1, \dots, R_N\}$ be the manipulators in the cell. We use *arm* to denote a serial-link manipulator with an independent controller. Arm R_i has n_i revolute joints and a fixed base transform $\mathbf{T}_i^{\text{base}} \in SE(3)$ with respect to the world frame W. The joint angles are the configuration variables of the system, $\Theta = \{\theta_1, \dots, \theta_{N_{\text{DOF}}}\}$, functions of time $\theta_k(t)$. The configuration of arm R_i at time t is $\mathbf{q}_i(t) \in \mathcal{Q}_i \subset \mathbb{R}^{n_i}$, bounded by joint position limits, with velocity and acceleration bounds $\dot{\mathbf{q}}_i^{\max}$, $\ddot{\mathbf{q}}_i^{\max}$. We write $\mathcal{C}_i$ for arm R_i's configuration space when referring to it as a factor of the composite space $\mathcal{C}_1 \times \cdots \times \mathcal{C}_N$ of Section IV-A; the two symbols denote the same set.

The arms are mounted so that their reachable volumes intersect; we refer to the intersection as the shared workspace. Arm R_i is represented for collision purposes by a set of P convex parts $\{\mathcal{P}_{i,p}\}_{p=1}^{P}$, each rigidly attached to a link frame. The inter-arm separation is the minimum distance between two arms' part surfaces,

$$d_{ij}(\mathbf{q}_i, \mathbf{q}_j) = \min_{p,\,r} \operatorname{dist}\big(\mathcal{P}_{i,p}(\mathbf{q}_i),\, \mathcal{P}_{j,r}(\mathbf{q}_j)\big), \tag{1}$$

zero when the parts touch or interpenetrate. A pair is clear when $d_{ij} \geq \varepsilon$ for a clearance gate $\varepsilon > 0$, evaluated on the true collision mesh, not a capsule or bounding-volume approximation. We use "exact" throughout this paper for this sense – exact at each sampled instant, not a swept-volume or continuous-time safety certificate.

Each arm is given a task-space goal $\mathbf{X}_i^{\text{goal}} \in SE(3)$, located in the shared workspace. The inverse kinematics of the arm admits several branches realizing $\mathbf{X}_i^{\text{goal}}$, so the goal configuration $\mathbf{q}_i^{\text{goal}}$ is itself unknown. Because every arm must reach its goal configuration and remain there, the assignment $\{\mathbf{q}_i^{\text{goal}}\}$ must be mutually clear,

$$d_{ij}\big(\mathbf{q}_i^{\text{goal}}, \mathbf{q}_j^{\text{goal}}\big) \geq \varepsilon, \qquad \forall\, i < j. \tag{2}$$

Condition (2) is necessary for the problem to admit a solution: the endpoints of every trajectory are pinned at the goal configurations, and an assignment violating it cannot be repaired by any subsequent choice of path or schedule.

Arm R_i executes over $[t_i, t_f]$. We write its trajectory as a path composed with a time parameterization, $\mathbf{q}_i(t) = \boldsymbol{\gamma}_i(\phi_i(t))$, with $\boldsymbol{\gamma}_i : [0,1] \to \mathcal{Q}_i$ and $\phi_i : [t_i, t_f] \to [0,1]$ non-decreasing, $\phi_i(t_i) = 0$, $\phi_i(t_f) = 1$. Both $\boldsymbol{\gamma}_i$ and ϕ_i are unknown. No relative motion constraint couples the arms, so the ϕ_i need not be the same for different i.

In isolation, arm R_i admits a preferred trajectory. Writing Π_i for the trajectories satisfying R_i's own endpoint, kinematic, and actuator constraints, and J_i for a per-arm cost such as execution time,

$$\boldsymbol{\gamma}_i^{\star} = \arg\min_{\boldsymbol{\gamma}_i \in \Pi_i} J_i(\boldsymbol{\gamma}_i). \tag{3}$$

The trajectories $\{\boldsymbol{\gamma}_i^{\star}\}$ are obtained independently and need not be jointly admissible: two such paths may violate (1) at some instant. We measure the departure of a trajectory from its isolated optimum as the largest joint-angle displacement the repair introduces anywhere along the path,

$$D_i = \max_{u \in [0,1]} \big\|\boldsymbol{\gamma}_i(u) - \boldsymbol{\gamma}_i^{\star}(u)\big\|_{\infty}, \tag{4}$$

reported in degrees, so that $D_i = 0$ exactly when the path is untouched and D_i is comparable across arms of differing joint count.

Problem. Given $\mathcal{R}$ with kinematic and actuator descriptions, collision geometry, and base placements in W; the task-space goals; and the gate ε; find goal configurations $\mathbf{q}_i^{\text{goal}}$ and trajectories $\theta_k(t)$ for all arms such that the total execution time is minimized, the departure of each arm from its isolated optimum is minimized subject to that, the goal configurations satisfy (2), the pairwise clearance

$$d_{ij}\big(\mathbf{q}_i(t), \mathbf{q}_j(t)\big) \geq \varepsilon, \qquad \forall\, t \in [t_i, t_f],\ \forall\, i < j, \tag{5}$$

holds throughout the motion, each arm reaches and holds its goal configuration, and joint limits are respected. The two criteria are ordered lexicographically, makespan first and total deviation second,

$$\operatorname{lex\,min} \ \Big(t_f - t_i\ ,\ \ \textstyle\sum_i D_i\Big), \tag{6}$$

so that $\sum_i D_i$ is minimized only among the schedules already attaining the minimum makespan.

IV. Approach

A. Coordination Domain

The formulation of Section III is a search over $\mathcal{C}_1 \times \cdots \times \mathcal{C}_N$, where $\mathcal{C}_i$ is arm R_i's own configuration space as defined in Section III. We instead compute the isolated optima (3) independently and reconcile them. Retiming changes ϕ_i and

leaves $\boldsymbol{\gamma}_i$ unchanged, so $D_i = 0$; deformation changes $\boldsymbol{\gamma}_i$ and $D_i > 0$.

For a pair (i, j), let $\mathcal{U}_{ij} = [0, 1]^2$ be the coordination domain and define the interference set

$$\Omega_{ij} = \Big\{(u, v) \in \mathcal{U}_{ij} : d_{ij}\big(\boldsymbol{\gamma}_i(u), \boldsymbol{\gamma}_j(v)\big) < \varepsilon\Big\}. \tag{7}$$

Ω_{ij} depends only on the paths. A schedule induces the curve $t \mapsto (\phi_i(t), \phi_j(t))$, monotone from $(0, 0)$ to $(1, 1)$ since both phases are non-decreasing; the schedule satisfies (5) for the pair iff the curve avoids Ω_{ij}.

Observation. Either a monotone path from $(0, 0)$ to $(1, 1)$ exists in $\mathcal{U}_{ij} \setminus \Omega_{ij}$, in which case a schedule exists with $D_i = D_j = 0$, or every monotone path meets Ω_{ij}, in which case no schedule satisfies (5) and at least one path must change. This is the coordination-diagram dichotomy already implicit in O'Donnell and Lozano-Pérez's two-robot construction [6]: a monotone curve fails to avoid a closed region exactly when that region separates $(0, 0)$ from $(1, 1)$ along every non-decreasing path – a statement about the shape of Ω_{ij} alone, not a new result requiring its own proof; we state it to fix the terminology ($\lambda^\star$, Eq. (8)) used in the rest of the paper.

Fig. 3(a) shows this domain for one interfering pair drawn from the six-arm cell trials, with the shaded Ω_{ij} and the resulting $\lambda^\star = 0.15$ marked; Fig. 3(b), in Section IV-E, shows the phase trajectories this pair actually executes under that bound.

Constraining every arm to $|\phi_i(t) - t/T_0| \le \lambda$ confines the coordination curve to the band $|u - v| \le 2\lambda$. Writing $\lambda^\star$ for the smallest such bound admitting a free monotone path over all pairs,

$$t_f - t_i \;\approx\; T_0\big(1 + \lambda^\star\big). \tag{8}$$

Equation (8) identifies $\lambda^\star$ as the entire cost of temporal resolution, which motivates searching λ in ascending order and is developed further in Section IV-E.

The method comprises six stages, summarized in Fig. 2 and Algorithm 1.

B. Stage 1: Goal Configuration Selection

Inverse kinematics is solved per arm from multiple seeds using sequential quadratic programming. Solutions are ranked by

$$\left(\max_k \frac{|\Delta q_k|}{\dot{q}_k^{\max}},\;\; \sum_k w_k |\Delta q_k|\right), \tag{9}$$

operation time first, weighted joint travel to break ties. A min-conflicts search over the resulting branches assigns one configuration per arm satisfying (2), tried first at a slack margin $\varepsilon + \delta$ and, failing that, at ε. An instance with no satisfying assignment is reported infeasible.

C. Stage 2: Seed Trajectory

Each arm is assigned a clamped cubic B-spline,

$$\boldsymbol{\gamma}_i(u) = \sum_{c=1}^{m} B_{c,3}(u)\, \mathbf{c}_{i,c}, \qquad u \in [0, 1], \tag{10}$$

Algorithm 1 Synchronized trajectory generation

Require: arms $\mathcal{R}$, starts $\mathbf{q}_i^0$, goals $\mathbf{X}_i^{\text{goal}}$, gate ε, nominal duration T_0
Ensure: schedule $\{\boldsymbol{\gamma}_i, \phi_i\}$ satisfying (5), or INFEASIBLE

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Solve IK per arm; select mutually clear goals via (2)
Generate a clamped cubic B-spline seed per arm; scale duration to limits
Scan all pairs on the exact mesh; flag interfering pairs
repeat
    Search λ ascending; for each, search per-pair leader orderings (Section IV-E)
    if a schedule clears all pairs then goto Execute
    end if
    Escape residual pairs jointly by cluster (Section IV-F)
until clear, or no further progress, or budget exhausted
Execute: pre-flight on exact mesh; stream commands; settle per (19)
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with the second and second-to-last control points duplicated onto the endpoints, giving zero endpoint velocity. Duration is scaled until velocity and acceleration limits hold for every joint; all arms are resampled onto a common clock.

D. Stage 3: Interference Detection

All $N(N-1)/2$ pairs are scanned along the seed trajectories on the exact collision geometry. Writing $\mathbf{o}_{i,p}$ and $\rho_{i,p}$ for the centre and radius of the sphere bounding part $\mathcal{P}_{i,p}$ – distinct from the control points $\mathbf{c}_{i,c}$ of (10) – a provable lower bound follows from the fact that each sphere contains its part,

$$\text{lb}_{ij} = \min_{p,r}\big(\|\mathbf{o}_{i,p} - \mathbf{o}_{j,r}\| - \rho_{i,p} - \rho_{j,r}\big) \;\le\; d_{ij}, \tag{11}$$

precedes each exact query; when $\text{lb}_{ij} > \varepsilon$ the pair is clear by construction and the exact mesh query is skipped.

E. Stage 4: Kuramoto Phase Synchronization

This stage implements contribution 2 (Section I) and is treated in full.

1) The Classical Model: The Kuramoto model [18] describes a population of N oscillators with phases $\theta_i \in S^1$ (the circle) evolving under

$$\dot{\theta}_i = \omega_i + \frac{K}{N}\sum_{j=1}^{N} \sin(\theta_j - \theta_i), \tag{12}$$

where ω_i is a natural frequency and $K > 0$ a coupling gain. For K above a critical value the population phase-locks: the sinusoidal coupling term drives θ_i toward the mean phase of its neighbours, and the population converges to a common, or fixed relative, phase [19]. This is the formalism underlying oscillator-based central pattern generators for locomotion [17], where the quantity being synchronized is a periodic gait phase and the coupling is attractive by design. Repulsive coupling, $K < 0$ in (12) or a per-oscillator sign as in the conformist/contrarian population of Hong and Strogatz [20], drives θ_i away from the mean phase instead; it is this

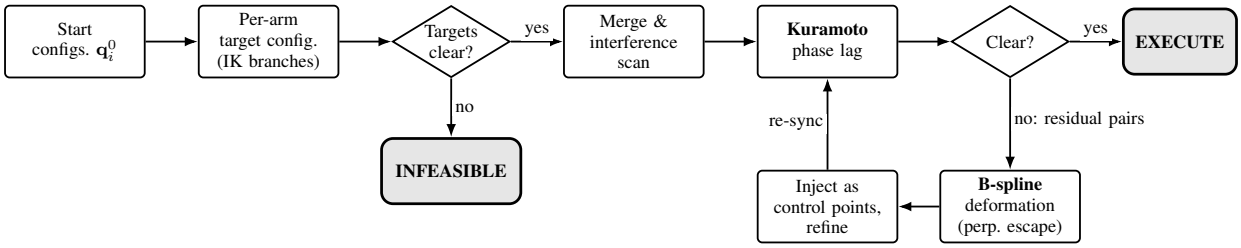


Fig. 2. Pipeline architecture. Independently planned trajectories are merged into the shared workspace and reconciled temporally where possible (Section IV-E); only the pairs that remain are deformed (Section IV-F), and every geometric change returns to the synchronizer.

qualitative behaviour, not the specific periodic formalism, that we adapt below.

2) Why the Classical Model Does Not Apply Directly: Three properties of (12) do not transfer to interference resolution among finite, point-to-point trajectories.

- *Domain.* $\theta_i \in S^1$ is periodic; our phase indexes progress along a single finite path and must terminate at $\phi_i = 1$ rather than circulate.
- *Sign of coupling.* Phase locking is convergence to a common phase, precisely the condition that produces Ω_{ij} in (7); the coupling required here is repulsive, driving phases apart, in the qualitative sense of [20] but on a terminating rather than a periodic domain.
- *Form of coupling.* The sinusoidal term in (12) is additive and can be negative – unchanged even for the contrarian oscillators of [20], which use a negative K_i in the same mean-field form. Applied to our $\phi_i \in [0,1]$, an additive term of either sign can drive $\dot{\phi}_i$ negative, i.e. an arm reversing along its own path, excluded by the monotonicity requirement of Section IV-A.

Table I summarizes the three departures.

TABLE I
CLASSICAL KURAMOTO COUPLING VERSUS THE COUPLING USED HERE

Property	Classical (12)	This paper
Phase domain	S^1, periodic	$[0,1]$, terminating
Coupling sign	attractive	repulsive
Coupling form	additive	multiplicative
Objective	common phase	staggered phase
Guarantee on rate	none	$\dot{\phi}_i \geq 0$ always

3) Multiplicative Rate Modulation: Each arm advances at

$$\dot{\phi}_i = \omega\, s_i, \qquad \omega = \frac{1}{T_0}, \qquad s_i \in [0,1]. \tag{13}$$

Because s_i is constrained to $[0,1]$ instead of added as a signed term, $\dot{\phi}_i \geq 0$ holds identically, for any s_i: monotonicity comes from this construction, not from tuning a gain. The coupling acts only through s_i,

$$s_i = \min_{j \in \mathcal{K}_i} \operatorname{clip}\left(1 - \frac{d_{\text{safe}} - d_{ij}}{d_{\text{safe}} - d_{\text{hard}}}, s_{\min}, 1\right), \tag{14}$$

evaluated only over $\mathcal{K}_i$, the arms Stage 3 flagged as interfering with arm i, and applied only when arm i is the trailing arm of the pair, $\phi_i < \phi_j$, with $d_{\text{safe}} > d_{\text{hard}}$. The rate is reduced continuously as the pair closes from d_{safe} to d_{hard} and recovers continuously as it reopens past d_{safe}; because $s_{\min} > 0$ (measured $s_{\min} \approx 0.22$, Table II), the trailing arm is never fully stopped, only slowed to about a fifth of nominal rate at the deepest point the band allows. Because s_i is a minimum over $\mathcal{K}_i$, an arm in conflict with several others yields to the most pressing – and because ϕ_i is a single running integral of s_i, lag an earlier conflict has already imposed on arm i carries forward into how it approaches its next one, rather than resetting between pairs.

4) Termination: Repulsive coupling alone does not guarantee that a braked arm ever finishes. We impose the floor

$$\phi_i(t) \geq \min\left(1, \frac{t}{T_0} - \lambda\right), \tag{15}$$

which bounds how far arm i may fall behind nominal progress and forces $\phi_i(t) \to 1$ as $t \to T_0(1+\lambda)$ regardless of how (14) behaves. Equations (13)-(15) together guarantee that every schedule produced by this stage terminates, independent of the interference geometry.

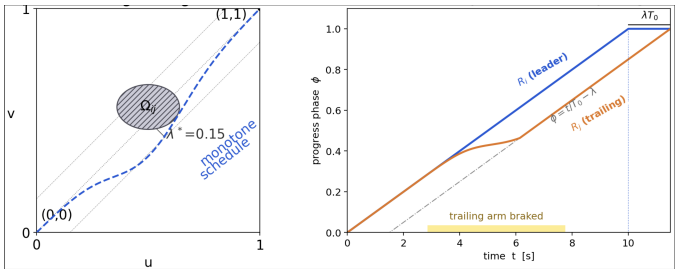


Fig. 3. A concrete instance from the six-arm cell trials, one interfering pair with R_i leading and R_j trailing. (a) The coordination domain $\mathcal{U}_{ij}$ with Ω_{ij} shaded and the accepted monotone schedule, giving $\lambda^\star = 0.15$. (b) The resulting phase trajectories under (13)–(15): R_j's rate is reduced while the pair is within the brake band (shaded interval), R_i passes unimpeded, and both arms reach $\phi = 1$ at the makespan predicted by (8), $T_0(1+\lambda^\star) = 10\,\text{s} \times 1.15 = 11.50\,\text{s}$, a 15.0% overhead directly set by $\lambda^\star$.

5) Ordering and Complexity: Which arm leads a given pair is not fixed by (14): with K interfering pairs there are 2^K leader assignments, realized by briefly holding the intended follower at its start; a single global bias is insufficient once different pairs require different leaders. All admissible orderings are tried per candidate λ, and every resulting schedule is verified against the exact gate (1) before acceptance – (14) decides when to brake inside the integration loop, never to

certify a result. Because $\mathcal{K}_i$ restricts coupling to flagged pairs, one integration step costs $O(K)$ rather than $O(N^2)$, and λ is searched ascending so the search accepts the cheapest schedule among the tested candidates, per (8) – cheapest relative to this finite search, not a certified global optimum.

F. Stage 5: B-Spline Deformation (Perpendicular Escape)

Pairs Stage 4 cannot separate are resolved by *deformation*: the seed path of each arm in the pair is displaced away from the contact by displacing its B-spline control points. The direction and magnitude of the displacement come from a perpendicular escape at the deepest contact; the displacement is then realized as a trajectory by the control-point update given later in this section (Eq. (18)). We use *deformation* throughout the paper, including in Table III, to name this combined stage; Fig. 4 shows the effect, and the remainder of this section describes first the escape direction and then its injection into the B-spline.

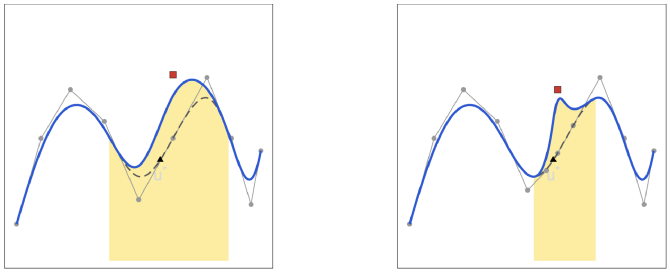


Fig. 4. Perpendicular escape injected as a B-spline deformation, before and after local refinement. In each panel the dotted curve is the unmodified nominal path and the solid curve is the path after the same physical displacement at $u^\star$ (marked) is pushed through the control polygon (Eq. (18)). (a) Without refinement, the edit is carried by the original control points alone and visibly perturbs 53% of the path. (b) After inserting control points local to $u^\star$, the identical displacement is confined to 23% of the path, leaving the rest of the trajectory exactly as planned.

For pairs unresolved by Stage 4, the deepest contact along the residual arc supplies witness points $\mathbf{p}_i, \mathbf{p}_j$ and separating direction $\hat{\mathbf{n}} = (\mathbf{p}_i - \mathbf{p}_j)/\|\mathbf{p}_i - \mathbf{p}_j\|$. Both arms are displaced,

$$\Delta\mathbf{q}_i = \alpha\, \mathbf{J}_i(\mathbf{p}_i)^\top \hat{\mathbf{n}}, \qquad \Delta\mathbf{q}_j = -\alpha\, \mathbf{J}_j(\mathbf{p}_j)^\top \hat{\mathbf{n}}. \tag{16}$$

where step size $\alpha > 0$ the closest points are unique, d is differentiable and, by Danskin's theorem [21], $\nabla_{\mathbf{q}_i} d = \mathbf{J}_i(\mathbf{p}_i)^\top \hat{\mathbf{n}}$, so

$$\langle \nabla_{\mathbf{q}_i} d, \Delta\mathbf{q}_i \rangle = \alpha \|\mathbf{J}_i(\mathbf{p}_i)^\top \hat{\mathbf{n}}\|^2 \geq 0, \tag{17}$$

an ascent step on separation, equality only when no joint upstream of the contact can displace it.

The displacement is mapped to control points by the minimum-norm solution of $\sum_c B_{c,3}(u^\star)\Delta\mathbf{c}_c = \Delta\mathbf{q}$,

$$\Delta\mathbf{c}_c = \frac{B_{c,3}(u^\star)\,\Delta\mathbf{q}}{\sum_j B_{j,3}(u^\star)^2}, \tag{18}$$

with endpoint control points excluded. The polygon is refined near $u^\star$ by knot insertion before the push, confining the deformation to the contacting arc. The step size is halved until the global violation count strictly decreases. Interfering pairs sharing an arm are repaired jointly. After every accepted change, Stage 4 is re-invoked: the deformation is local to the contacting pair and arc, but its effect on Ω_{ik} for a third arm k sharing a base with either deformed arm is not, so the correctness of a local geometric edit is not established by the local edit alone. Re-invoking the full Kuramoto sweep rather than a targeted check closes this gap: it verifies, and if necessary absorbs, any interference the repair newly introduces elsewhere, without a global re-optimization over all trajectories. Section V reports how often this second invocation is what actually resolves the scene.

G. Stage 6: Execution

Trajectories are streamed as joint position commands. A pre-flight check refuses any schedule violating (5) on the exact geometry. Settling is monitored to convergence with a window sized from a first-order model of the position loop,

$$t_{\text{settle}} = \tau \ln(e_0/e_{\text{tol}}), \qquad e_0 \approx v_{\text{peak}}\tau. \tag{19}$$

V. Results

A. Experimental Setup

We evaluate the method on cells of two, four, and six arms, all placed on the same 1.39 m base spacing so that ensemble size is the only quantity varying across the three configurations. We take 1.39 m as the SAFE MINIMUM base spacing for this manipulator: at this distance, an arm at its resting (home) configuration does not intersect the reachable volume of a neighbouring arm, so any subset of the cell can be commanded to move while the remaining arms stay at rest with no collision between an active arm and an idle one. This property is what makes a decoupled, per-subset mode of operation possible in the first place; spacing closer than 1.39 m was not evaluated, since it would put a resting arm inside an active neighbour's working volume and violate this guarantee.

For each ensemble size $N \in \{2, 4, 6\}$, we attempted $n = 150$ independent trials. The goal $\mathbf{X}_i^{\text{goal}}$ for arm R_i was sampled independently and uniformly by volume from the shared workspace $W_N = \bigcap_{i=1}^{N} W_i$ (Section III), where W_i is arm i's reachable workspace: points were drawn from W_N's axis-aligned bounding box and rejected until inside W_N. Goal configurations were selected by the solver per the ranking of Eq. (9). No reachability pre-filter was applied before a goal reached Stage 1. A trial was classified IK_FAIL if some goal admitted no IK solution, ENDPOINT_CSP_FAIL if Stage 1's endpoint CSP (Eq. (2)) found no mutually clear assignment, or STAGE_ERROR on an unhandled later fault. All three are retained in the $n = 150$ denominator rather than discarded and re-drawn, so $n = 150$ is trials *attempted*, not accepted after filtering – which keeps the "interfering trials" count in Table III interpretable as a rate over a well-defined population. Table II summarizes the fixed parameters, and Fig. 5 shows the resulting 2/4/6-arm cells during live execution.

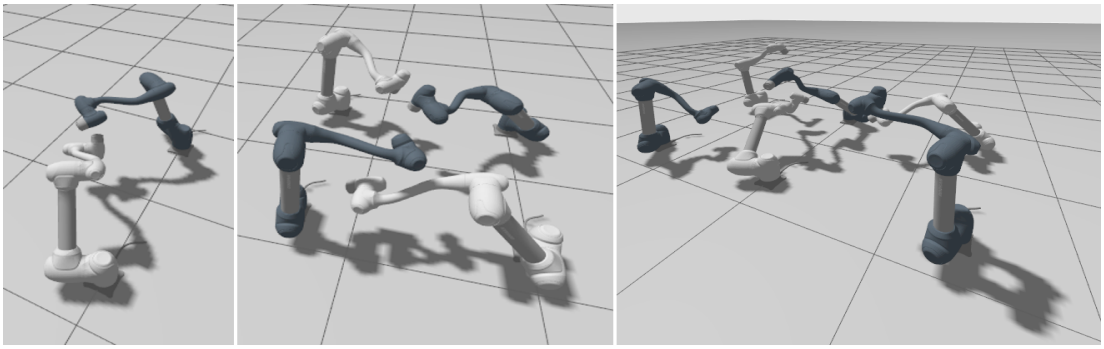


Fig. 5. The 2/4/6-arm cells rendered in the collision-mesh viewer, (a) two arms, (b) four arms, (c) six arms – the physical scale-up underlying the resolution outcomes of Table III. Panel (c) is shown from a different camera pose than (a)-(b) to keep all arms visible.

TABLE II
EXPERIMENTAL CONFIGURATION

Item	Value
Manipulators	Doosan M1013, 6 DOF each
Ensemble sizes	2, 4, 6 arms
Base spacing (all cells)	1.39 m (safe minimum)
Collision geometry	10 convex parts per arm
Clearance gate ε	0.05 m, surface to surface
Nominal duration T_0	10 s
Command rate	100 Hz
End-effector tolerance	10 mm
Stage 5 iteration cap	25 escape iterations/cluster
Stage 5 time budget	300 s wall-clock
Minimum brake rate $s_{\min}$	0.22 (never a full stop)

TABLE III
RESOLUTION OUTCOME BY ENSEMBLE SIZE (N = 150 PER CELL)

Quantity	2 arms	4 arms	6 arms
Pairs $N(N-1)/2$	1	6	15
Interfering trials	29	72	85
Resolved by Kuramoto alone	10.3%	11.1%	12.9%
Resolved by deformation+Kuramoto	69.0%	65.3%	67.1%
Unresolved	20.7%	23.6%	20.0%
Planning success	96.0%	88.7%	87.3%
Mean exec. time, interfering	10.8 s	12.6 s	14.0 s
Mean overhead, interfering	+8%	+26%	+40%

B. Resolution Outcome

The two rows above labelled "Resolved by Kuramoto alone" and "Resolved by deformation+Kuramoto" both invoke Stage 4, not two independent mechanisms in parallel. *Kuramoto alone* means phase synchronization by itself satisfies (5), so no path changes and every arm's deviation is zero. *Deformation+Kuramoto* means Stage 4 could not separate at least one pair on the unmodified paths, so Stage 5 deforms that pair specifically – never the whole ensemble – after which Stage 4 is re-invoked on the modified paths, the re-invocation mechanism already described in Section IV-F. The planning-success rate is therefore better read as the fraction of interfering trials Stage 4 resolves *eventually* than as two competing rates to be added. Planning success counts every trial that ends collision-free over all 150 attempted: the non-interfering trials plus the interfering ones the two tiers resolved. Its complement is the unresolved interfering trials together with the IK_FAIL and ENDPOINT_CSP_FAIL trials retained in the denominator (Section V). The six-arm cell is the only one where the two differ: its 17 unresolved trials alone would give 88.7%, and the two goal sets that admitted no IK solution account for the remaining gap to 87.3%.

Interference grows with ensemble size, from 19.3% of trials at two arms to 56.7% at six, tracking pair count. The *relative* split within the interfering set stays comparatively stable: deformation with re-synchronization accounts for 65–69% of interfering trials and Kuramoto alone for 10–13%, whether there are 29 or 85 such trials. The unresolved fraction is likewise stable at 20–24%, so the method's per-conflict resolution rate does not degrade disproportionately as interference load grows; what changes with N is how much interference there is, not how well each instance resolves.

Stage 4 (Kuramoto) is a retiming search that always terminates, by (15), with *some* schedule. It never itself fails to produce an answer. What can fail is Stage 5, when it is reached. The residual pairs it inherits are ones where the two paths are already mutually entangled instead of simply crossing: both arms' swept volumes wind around the shared region in the same rotational sense instead of approaching from opposite sides, so displacing either arm's control points perpendicular to the current contact does not monotonically increase separation the way it does for a transversal crossing. The witness point can simply relocate to a new contact a comparable distance away, and the escape spends its iterations chasing it instead of clearing it. Stage 5 is a bounded local search, not a complete algorithm: at most 25 escape iterations per cluster and an overall wall-clock budget of 300 s (Section IV-F). A trial in the entangled regime can exhaust this budget before converging even though no proof exists that the pair is geometrically inseparable under a larger budget. "Unresolved" is therefore a property of Stage 5's allotted geometric search, not evidence that the Stage 4 retiming mechanism broke down. It is a direction for future work: detecting this entangled regime early (e.g. from the sign pattern of successive witness-point displacements) could route it directly to a more expensive but convergent repair instead of spending the ordinary escape budget on it first.

Measured execution time in Gazebo (Table III): mean time

among interfering trials rises from 10.8 s at two arms to 12.6 s at four and 14.0 s at six, a makespan overhead of $+8\%$, $+26\%$, and $+40\%$ over the nominal $T_0 = 10$ s, averaged only over the trials that actually interfered (29, 72, and 85 respectively), not diluted by the clean majority. Most of this overhead comes from Stage 4: braking a trailing arm's progress directly stretches the schedule, per Eq. (8). Stage 5 deformation contributes only slightly, since it changes path geometry instead of timing, and a longer deformed path costs a little extra traversal time at unchanged joint rates. The growth with N tracks pair count more than interference rate alone: a two-arm cell has only one possible pair, so an interfering trial there braked for at most one conflict, while a six-arm cell's 15 pairs make it more likely that an interfering trial's arm braked for several conflicts in sequence, and because ϕ_i is a single running integral of s_i (Section IV-E), that lag compounds within the trial rather than resetting between pairs.

C. Symmetric Escape Operator

TABLE IV
TWO-SIDED VERSUS ONE-SIDED ESCAPE ON A 1.62 CM CONTACT

Variant	Arm i	Arm j	Total	Iter.	Cleared
Two-sided (proposed)	2.35°	1.08°	3.43°	1	yes
One-sided	5.15°	0°	5.15°	2	yes

Table IV shows the two-sided displacement halves the deviation incurred by the more heavily loaded arm; on a deep proximal contact the one-sided variant failed to clear at all, because moving a single arm dragged its own distal links into a new contact. This two-sided property, together with the local refinement and mandatory re-synchronization already described in Section IV-F, is why the deformation rate in Table III reflects many small, locally-confined, individually-verified edits rather than a few large ones.

VI. CONCLUSION

We presented an approach to generate collision-free synchronized configuration-space trajectories for an ensemble of serial-link manipulators sharing a common workspace, resolving inter-arm interference temporally before geometrically: each arm advances along its unmodified nominal B-spline under a Kuramoto phase-oscillator model, adapted from its classical periodic, attractive, additive form into a non-periodic, repulsive, multiplicative one, and only pairs phase modulation cannot separate are deformed – a perpendicular escape at the deepest contact, injected as B-spline control points with local refinement, with the full synchronization sweep re-run after every geometric change. Equal and opposite displacements to both arms of a colliding pair roughly halve the deviation each arm incurs versus a fixed-direction prior and succeed where a one-sided escape stalls, with all collision decisions made on the manipulators' true convex collision meshes.

The method assumes goal configurations are mutually collision-free; interfering goals must return to the inverse-kinematics branch-selection stage. Escape leverage is also not uniform along the kinematic chain, since only joints upstream of a contact can displace it. Future work includes physical hardware validation and redundant manipulators.

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